

Refrigeration & Air-Conditioning

Refrigeration - A process in which heat is rejected & low temperature is maintained.

Refrigerant :- A substance which absorbs the heat & maintain the low temp.

(A) Primary - Such refrigerant cooled by itself.

ex - R-12, R-22, R-134a, NH₃ etc.

(B) Secondary - not cooled by itself.

they get coolness by another and then getting cooled & they cool the another thing.

ex. NaCl

Refrigeration Cycle:- Cycle in which refrigeration process is taken place, in which heat is rejected & low temp is maintained.

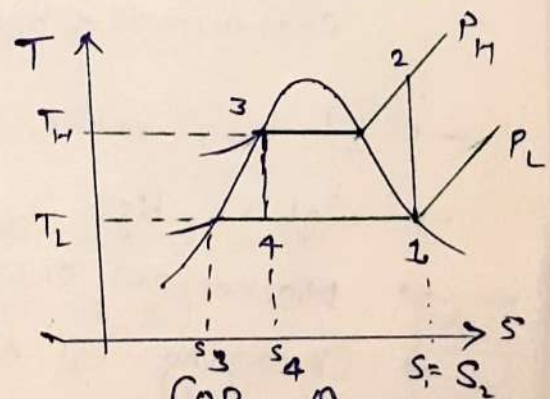
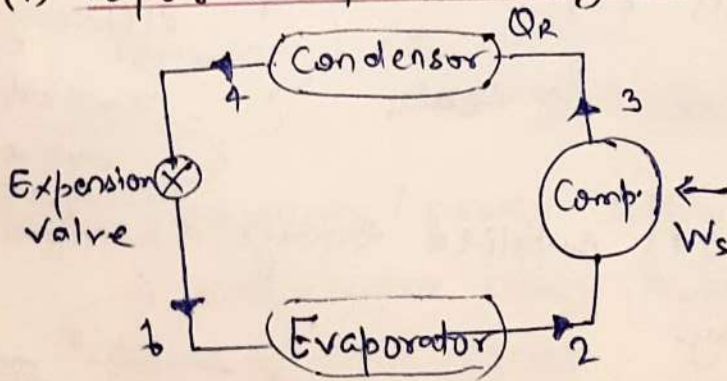
ex VCR → Vapor Compression cycle

VAR → Vapor Absorption Cycle

unit → Ton of Refrigeration (TR)

Cop ⇒ Coefficient of performance

(i) Vapor Compression cycle



$$COP = \frac{Q_a}{W_D}$$

(A) Compressor ⇒ Increase temp & pressure

(B) Condensor ⇒ Reject latent heat and convert high vapor into liquid form.

(10)

Expansion valve : reduce pressure & temp of refrigerant.

Evaporator : Evaporates liquid refrigerant, no heat is absorbed or rejected.

Working

- Compression process $\rightarrow$ get vapour form, increase temp & pressure. $W = h_2 - h_1$
- Condensing process $\rightarrow$ Reject the latent heat from high pressure & temp. vapour refrigerant. Condense into liquid form. (phase will change) (Low temp, High Press)
- Expansion valve $\rightarrow$ through mm hole, particle will pass and pressure & temp will decrease in liquid form. (No latent heat absorbed.) (with some vapour)
- vapourizing process $\rightarrow$ liquid + vapour is evaporated and change into vapour. at constant pressure & temp. (High)

Isentropic process $\rightarrow$ 1 to 2 & 3 to 4
condensor process 2 to 3
evaporated process 4 to 1

$$\text{COP} = \frac{1}{\text{Efficiency}} = \frac{\text{Heat extracted}}{\text{work done}}$$

$\rightarrow$ It has more wear & tear.

$\rightarrow$ COP is high

$\rightarrow$ Mechanical energy is supplied through a compressor

$\rightarrow$ Charging is easy.

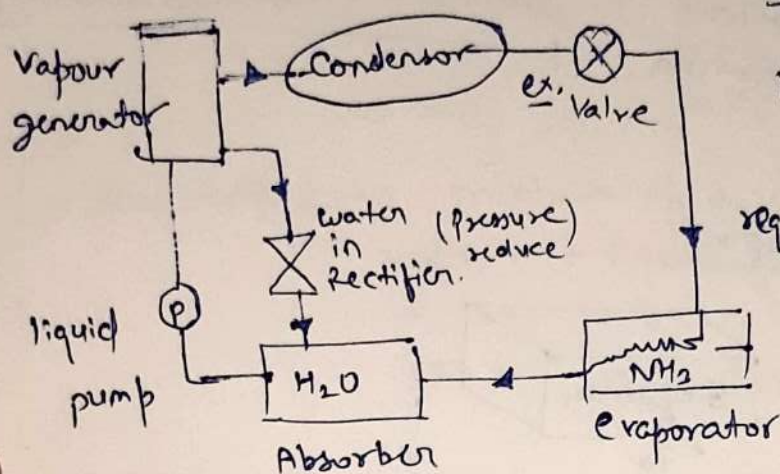
* 1 TR = $1000 \times 335 \text{ kJ in 24 hour}$ $\times 335 \text{ kJ} \Rightarrow$ latent heat.

= cooling effect by melting of 1000 kg of ice at 0°C temp in 24 hr

1 TR = 210 kJ/min = 3.5 kW

iii) Vapour Absorption Refrigeration cycle (VAR)

(11)



→ Used in Ice making Plants, less wear & tear.

Advantage

liquid is compressed

required input work is less

→ expensive.

→ complex and space consuming.

→ less efficient for large plant.

→ Cop is less.

→ Heat energy is utilized.

→ Charging is difficult.

Absorber → Absorbs the refrigerant (H_2O)

Condensor → Convert vapour to liquid form.

Vapour generator → Used to generate the heat

liquid pump → transfer the fluid from one place to another part.

generator → generate

pressure reducing valve → which reduce pressure & convert in liquid.

expansion valve → reduce pressure & temp of refrigerant.

propent at low temp $NH_3 + H_2O$ mix

at high temp → NH_3 & H_2O are separate.

* Working → work / power supply to liquid pump.

→ evaporator (NH_3), Absorber (H_2O)

* (Aqua ammonia cycle)

→ Heat supply to generator $H_2O + NH_3$ separate
(liquid) (Vapor)

→ Pressure reducing valve consume hot water and ammonia will collect in Condensor (weak solution)
(Rich)

→ Condensor convert it in liquid.

NH_3
H_2O

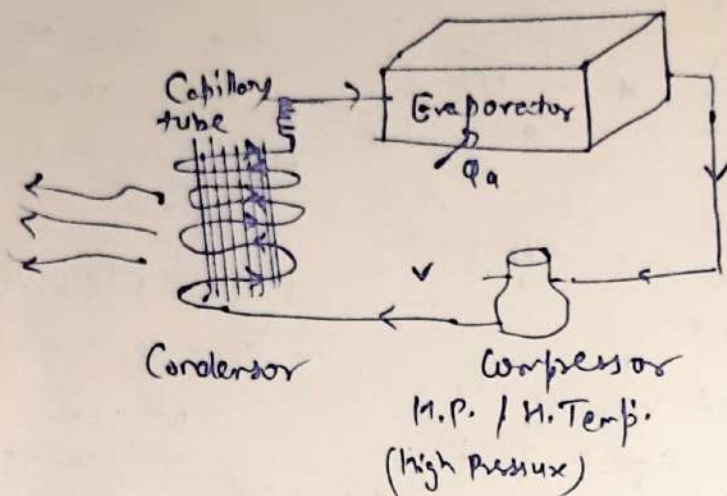
• Nomenclature of Refrigerant

$\text{CaH}_b\text{F}_c\text{Cl}_d$ = chemical formula.

$(b+c+d = 2a+2)$

Name = $\text{R}(a-1)(b+c)c$

Domestic Refrigerator



• Application of Refrigeration

- In water cooler to supply cold water.
- To produce ice.
- For preservation of food, veg., ice cream, medicines, blood, tissues.
- Industrial application.
- AC in Houses, offices, hospitals.

Desirable Property of Refrigerant

- Low boiling point, freezing point high Q_a
- Easy to liquify.
- Chemical stability.
- Odourless, no hazardous effect.
- Low cost
- Non flammable

Types of Refrigerant

(19)

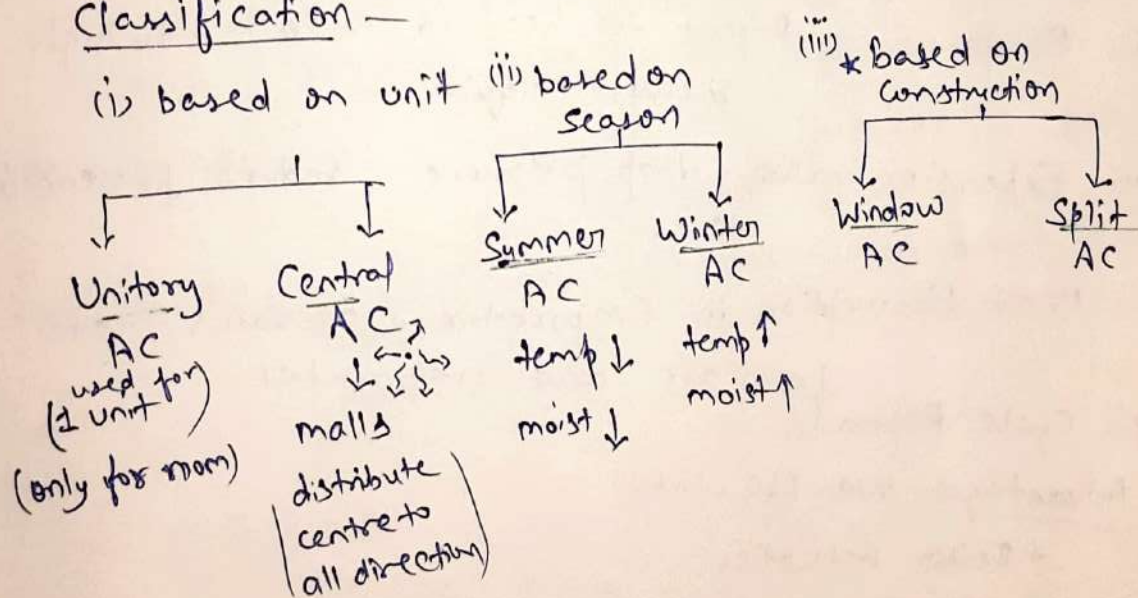
(P3)

- Halocarbons (Freons) → CFC → R11, R12, R113
HCFC → R22, R123
HFC → R134a
- Azeotropic → mixture of two or more refrigerants vapour + liquid form.
R-502 ⇒ 8.8% R22 + 91.2% R115
R-503 ⇒ 40.1% R23 + 59.9% R12
- zeotropic → mixture of composition in liquid phase is different.
- Inorganic → CO₂, NH₃, H₂O etc
- Hydrocarbon → R170, R290 etc. both industry & domestic.

* Air Conditioning System

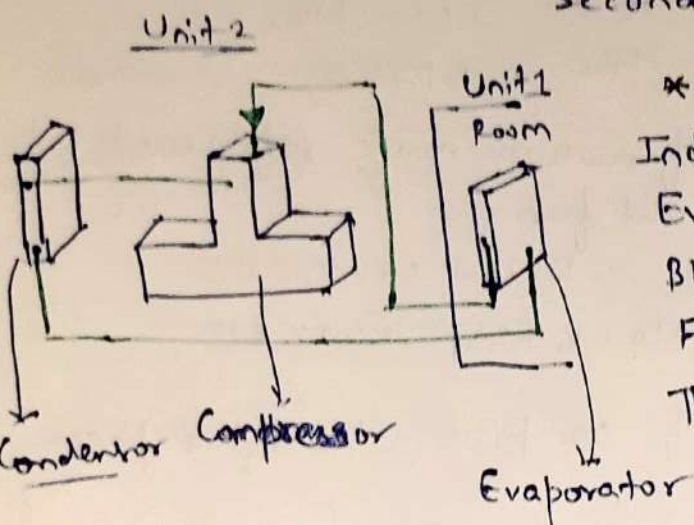
- Temperature is maintained.
- Dust particle are removed
- Moisture & Humidity is maintained.

Classification —



(A) Split AC → Consists two parts (Unit)

One indoor unit
Second → outdoor unit. } Connected by pipe & wire.



* Components

Indoor
Evaporator
Blower fan
Filters
Thermostat

Outdoor
Compressor
Condenser coil
Fan
Expansion valve

Working based on VCR cycle. transfer heat from the indoor air to outside environment.

Steps

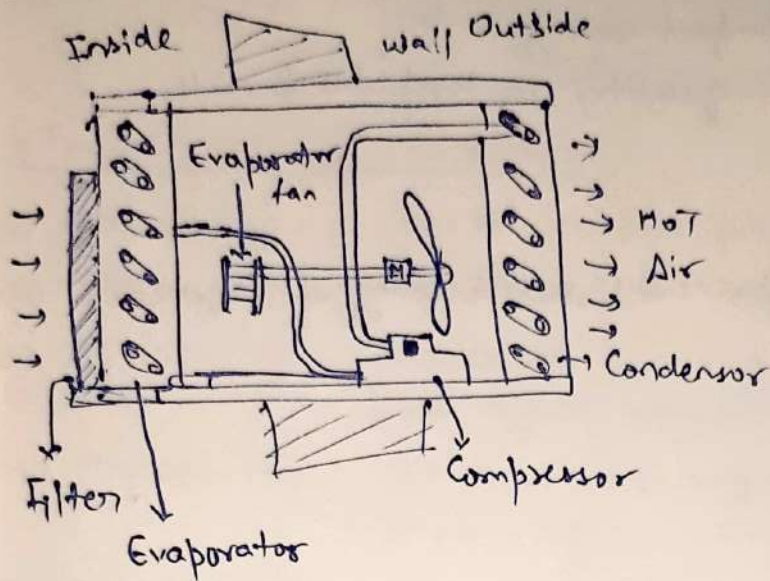
- (i) Thermostat senses the temp.
temp of room $>$ then AC turns on.
- (ii) Compressor compresses low pressure gas into high high temp & high pressure gas.
- (iii) Condensor → Reject the heat & convert into high pressure liquid.
- (iv) Expansion valve drop pressure, reduce pressure & temp.
- (v) Heat Absorption in Evaporator → absorbs heat from air and evaporates.
- (vi) Cycle Repeats.

Advantage over (AC window)

- Better Aesthetics
- efficient cooling of larger room.
- More installation flexibility.

at Steel

(A) * Window A C



* Parts

- Evaporator Fan
- Evaporator
- Expansion valve
- Filter
- Condenser
- Compressor

* A window AC is a compact, self-contained unit designed to cool a single room.

It is typically mounted in a window or a hole in an external wall.

working

based on refrigeration cycle, which involves heat absorption & release using a refrigerant.

1. Thermostat detects temperature -
2. Compressor starts → compresses the low pressure gas into a high pressure, high temp gas.
3. Heat release at the condenser → A fan blow outside air over the coil & refrigerant loses heat to outside, becoming becoming high pressure liquid.
4. Expansion valve → reduce the pressure & temp. of liquid.
5. Heat absorption at evaporator → Refrigerant absorbs heat from the air & evaporates.
6. cycle repeats - - -

Disadvantage

- (i) Noise
- (ii) Cooling effect.
- (iii) Installation Difficult.

Application of AC

- Homes & offices → comfort cooling.
- Maintain indoor air quality in Hospital & malls.
- in Vehicles.
- Factories & labs.
- Data centers to prevent overheating of servers.